

SATIM Pay Full-Stack Web Application

Developer Usage Report and Documentation

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Executive Summary

This comprehensive documentation provides developers with complete instructions for deploying, configuring, and maintaining the SATIM Pay full-stack web application. The application represents a modern banking platform built with cutting-edge technologies including FastAPI, Next.js, MongoDB, and AI-powered chatbot functionality.

The system has been successfully integrated and tested, with a fully operational backend API, responsive frontend interface, populated database with realistic dummy data, and functional AI chatbot. All components communicate seamlessly, providing a robust foundation for a production-ready banking application.

This report covers every aspect of the application from initial setup to advanced configuration, ensuring developers can quickly understand, deploy, and extend the system according to their specific requirements.

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System Architecture Overview

The SATIM Pay application follows a modern three-tier architecture designed for scalability, maintainability, and security. The system consists of three primary components that work together to deliver a comprehensive banking experience.

Architecture Components

Frontend Layer (Presentation Tier) The frontend is built using Next.js 15.2.4 with TypeScript, providing a modern, responsive user interface. The application utilizes Tailwind CSS for styling and incorporates Radix UI components for consistent, accessible user interface elements. The frontend communicates with the backend through RESTful API calls and maintains state management for user sessions and application data.

Backend Layer (Application Tier) The backend is implemented using FastAPI, a modern Python web framework that provides automatic API documentation, request validation, and high performance. The backend handles business logic,

authentication, data processing, and external service integrations including the AI chatbot functionality through the Groq API.

Database Layer (Data Tier) MongoDB serves as the primary database, providing flexible document storage that accommodates the complex data structures required for banking operations. The database includes collections for users, accounts, transactions, and notifications, with proper indexing for optimal performance.

Technology Stack

Backend Technologies: - FastAPI 0.115.13 for API development - Python 3.11 runtime environment - Motor 3.7.1 for asynchronous MongoDB operations - PyMongo 4.13.2 for database connectivity - Passlib with bcrypt for password hashing - Python-JOSE for JWT token management - Groq API for AI chatbot functionality - Uvicorn for ASGI server implementation

Frontend Technologies: - Next.js 15.2.4 for React framework - TypeScript for type safety - Tailwind CSS 3.4.17 for styling - Radix UI components for accessibility - Lucide React for iconography - React Hook Form for form management - Zod for schema validation

Database and Infrastructure: - MongoDB 7.0.21 for document storage - Ubuntu 22.04 LTS operating system - Node.js 20.18.0 for frontend build tools - npm/pnpm for package management

Data Flow Architecture

The application follows a unidirectional data flow pattern where user interactions in the frontend trigger API requests to the backend, which processes the requests, interacts with the database as needed, and returns structured responses. Authentication is handled through JWT tokens that are stored securely in the browser's localStorage and included in subsequent API requests.

For real-time features like notifications, the system is designed to support WebSocket connections, though the current implementation uses polling for simplicity. The AI chatbot integrates seamlessly into this flow by processing user messages through the backend API, which then communicates with the Groq AI service to generate intelligent responses.

Prerequisites and Dependencies

Before beginning the installation process, developers must ensure their development environment meets the minimum system requirements and has all necessary dependencies installed. This section provides detailed information about the required software, versions, and configuration steps.

System Requirements

Operating System Compatibility: The SATIM Pay application has been developed and tested on Ubuntu 22.04 LTS, but it is compatible with most modern Linux distributions, macOS, and Windows systems. For production deployments, Ubuntu Server 22.04 LTS or CentOS 8+ is recommended for optimal performance and security.

Hardware Requirements: - Minimum 4GB RAM (8GB recommended for development)
- 20GB available disk space - Multi-core processor (2+ cores recommended) - Stable internet connection for package downloads and API integrations

Required Software Dependencies

Python Environment: The backend requires Python 3.11 or higher. While Python 3.9+ may work, Python 3.11 is specifically recommended for optimal compatibility with all dependencies. Developers should install Python using their system's package manager or from the official Python website.

Node.js Environment: The frontend requires Node.js 20.18.0 or higher. This version includes npm and provides the necessary JavaScript runtime for building and running the Next.js application. Developers can install Node.js from the official website or use a version manager like nvm for easier version management.

Database System: MongoDB 7.0.21 is required for data storage. The application uses MongoDB's document-based storage for flexibility in handling complex banking data structures. MongoDB can be installed locally for development or configured to connect to a remote MongoDB instance or MongoDB Atlas for production use.

Package Managers: - pip3 for Python package management - npm or pnpm for Node.js package management - Optional: yarn as an alternative to npm

Development Tools

Code Editors and IDEs: While any text editor can be used, the following are recommended for optimal development experience: - Visual Studio Code with Python and TypeScript extensions - PyCharm Professional for backend development - WebStorm for frontend development

Version Control: Git is required for version control and collaboration. The project includes a .gitignore file configured for both Python and Node.js development.

API Testing Tools: - Postman or Insomnia for API testing - curl for command-line API testing - Browser developer tools for frontend debugging

Environment Setup Verification

Before proceeding with installation, developers should verify their environment by running the following commands:

```
# Check Python version
python3 --version

# Check Node.js version
node --version

# Check npm version
npm --version

# Check MongoDB installation
mongod --version

# Check Git installation
git --version
```

All commands should return version numbers matching or exceeding the minimum requirements specified above.

Installation and Setup

This section provides step-by-step instructions for setting up the SATIM Pay application from initial download through complete configuration. The process is designed to be straightforward while ensuring all components are properly configured for development and testing.

Project Structure Overview

The project is organized into distinct directories for backend and frontend components:

```
Datalith-HAICK25-adlen-s-branch/
├── backend/
│   ├── app/
│   │   ├── core/
│   │   ├── models/
│   │   ├── routers/
│   │   └── services/
│   ├── requirements.txt
│   ├── .env
│   └── setup_database.py
├── frontend/
│   ├── app/
│   ├── components/
│   ├── lib/
│   ├── public/
│   ├── package.json
│   └── .env.local
└── README.md
```

Step 1: Project Extraction and Initial Setup

Begin by extracting the project files to your desired development directory. The project should be extracted to a location with appropriate read/write permissions for your development user.

```
# Navigate to your development directory
cd /path/to/your/development/folder

# Extract the project (if from zip file)
unzip Datalith-HAICK25-adlen-s-branch.zip

# Navigate to the project directory
cd Datalith-HAICK25-adlen-s-branch
```

Step 2: Database Installation and Configuration

MongoDB must be installed and configured before setting up the application components. The following instructions cover installation on Ubuntu systems, but similar steps apply to other operating systems.

MongoDB Installation on Ubuntu:

```
# Import MongoDB public GPG key
wget -qO - https://www.mongodb.org/static/pgp/server-7.0.asc | sudo apt-key add
-

# Add MongoDB repository
echo "deb [ arch=amd64,arm64 ] https://repo.mongodb.org/apt/ubuntu
jammy/mongodb-org/7.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-
org-7.0.list

# Update package database
sudo apt update

# Install MongoDB
sudo apt install -y mongodb-org

# Start MongoDB service
sudo systemctl start mongod

# Enable MongoDB to start on boot
sudo systemctl enable mongod

# Verify MongoDB is running
sudo systemctl status mongod
```

MongoDB Configuration: The default MongoDB configuration is suitable for development use. For production deployments, additional security configurations should be implemented, including authentication, SSL/TLS encryption, and network access restrictions.

Step 3: Backend Setup and Configuration

The backend setup involves installing Python dependencies, configuring environment variables, and initializing the database with dummy data.

Install Backend Dependencies:

```
# Navigate to backend directory
cd backend

# Install Python dependencies
pip3 install -r requirements.txt

# Install additional required packages
pip3 install motor pymongo bcrypt python-jose[cryptography] passlib[bcrypt]
```

Environment Configuration: The backend uses environment variables for configuration. Review and modify the `.env` file as needed:

```
# View current environment configuration
cat .env

# Edit environment variables if needed
nano .env
```

The `.env` file contains the following key configurations: - `GROQ_API_KEY` : API key for AI chatbot functionality - `MONGODB_URL` : MongoDB connection string - `DATABASE_NAME` : Name of the MongoDB database - `SECRET_KEY` : JWT token signing key (change for production)

Database Initialization: Run the database setup script to create collections and populate with dummy data:

```
# Run database setup script
python3 setup_database.py
```

This script creates the following: - User accounts (1 admin + 3 regular users) - Bank accounts with realistic balances - Transaction history - Notification records - Proper database indexes for performance

Step 4: Frontend Setup and Configuration

The frontend setup involves installing Node.js dependencies and configuring environment variables for API communication.

Install Frontend Dependencies:

```
# Navigate to frontend directory
cd ../frontend

# Install Node.js dependencies
npm install --legacy-peer-deps
```

Note: The `--legacy-peer-deps` flag is used to resolve dependency conflicts with React 19 and some UI component libraries.

Frontend Environment Configuration: Create or verify the `.env.local` file for frontend environment variables:


```
# Check if .env.local exists
cat .env.local

# Create if it doesn't exist
echo "NEXT_PUBLIC_API_URL=http://localhost:8000" > .env.local
```

Step 5: Application Startup

With all components configured, start both the backend and frontend servers.

Start Backend Server:

```
# From the backend directory
cd ../backend

# Start the backend server
python3 standalone_main.py
```

The backend server will start on `http://localhost:8000` and provide API endpoints for the frontend.

Start Frontend Server:

```
# From the frontend directory (in a new terminal)
cd frontend

# Start the frontend development server
npm run dev
```

The frontend server will start on `http://localhost:3000` and automatically open in your default browser.

Step 6: Verification and Testing

After both servers are running, verify the installation by accessing the application and testing key functionality.

Access Points: - Frontend Application: `http://localhost:3000` - Backend API Documentation: `http://localhost:8000/docs` - Backend Health Check: `http://localhost:8000/health`

Quick Functionality Test: 1. Open the frontend application in your browser 2. Verify the homepage loads correctly 3. Test the floating chatbot by clicking the chat icon 4.

Navigate to the authentication page 5. Attempt login with admin credentials (admin@satimpay.com / admin123)

Database Configuration

The SATIM Pay application uses MongoDB as its primary database system, chosen for its flexibility in handling complex financial data structures and its excellent performance characteristics for read-heavy banking applications. This section provides comprehensive information about database setup, schema design, and management procedures.

Database Architecture and Design

Database Structure: The application uses a single MongoDB database named `satim_pay_test` for development and testing purposes. In production environments, this should be changed to `satim_pay` or another appropriate name. The database is organized into four primary collections, each serving specific functional requirements of the banking application.

Collection Overview:

The `users` collection stores comprehensive user account information including personal details, security settings, and preferences. This collection serves as the central repository for user identity and authentication data, with proper indexing on email and username fields for efficient lookups.

The `accounts` collection maintains financial account information linked to users through `user_id` references. Each account includes balance information, account type classification, and transaction limits. This separation allows users to have multiple accounts while maintaining data integrity.

The `transactions` collection records all financial transactions within the system, including transfers, payments, and other banking operations. Each transaction includes comprehensive metadata for audit trails and regulatory compliance requirements.

The `notifications` collection manages user notifications for various system events, transaction alerts, and promotional messages. This collection supports different

notification types and delivery channels for comprehensive user communication.

Database Schema Details

Users Collection Schema:

```
{
  _id: ObjectId,
  email: String (unique, indexed),
  username: String (unique, indexed),
  password_hash: String,
  first_name: String,
  last_name: String,
  phone: String,
  role: String (enum: "user", "admin", "support"),
  is_active: Boolean,
  is_verified: Boolean,
  created_at: Date,
  updated_at: Date,
  profile: {
    avatar_url: String,
    bio: String,
    preferences: {
      language: String,
      currency: String,
      notifications: Boolean
    }
  },
  security: {
    login_attempts: Number,
    last_login: Date,
    account_locked: Boolean,
    two_factor_enabled: Boolean
  }
}
```

Accounts Collection Schema:

```
{
  _id: ObjectId,
  user_id: ObjectId (indexed),
  account_number: String (unique, indexed),
  account_type: String (enum: "checking", "savings", "business"),
  balance: Number,
  currency: String,
  status: String,
  created_at: Date,
  updated_at: Date,
  limits: {
    daily_transfer: Number,
    monthly_transfer: Number,
    daily_withdrawal: Number
  }
}
```

Transactions Collection Schema:

```
{
  _id: ObjectId,
  from_account: ObjectId (indexed),
  to_account: ObjectId (indexed),
  amount: Number,
  currency: String,
  type: String,
  status: String,
  description: String,
  reference: String,
  created_at: Date (indexed),
  completed_at: Date,
  metadata: {
    ip_address: String,
    user_agent: String,
    device_id: String
  }
}
```

Notifications Collection Schema:

```
{
  _id: ObjectId,
  user_id: ObjectId (indexed),
  title: String,
  message: String,
  type: String,
  status: String,
  created_at: Date (indexed),
  read_at: Date,
  metadata: {
    transaction_id: String,
    amount: Number,
    currency: String
  }
}
```

Database Initialization and Dummy Data

The application includes a comprehensive database setup script (`setup_database.py`) that creates all necessary collections, establishes proper indexing, and populates the database with realistic dummy data for development and testing purposes.

Dummy Data Overview:

The setup script creates one administrative user and three regular users, each with complete profiles and associated bank accounts. The administrative user has elevated

privileges for system management, while regular users represent typical banking customers with varying account balances and transaction histories.

User Accounts Created: - Administrative User: admin@satimpay.com (password: admin123) - Regular User 1: john.doe@example.com (password: password123) - Regular User 2: jane.smith@example.com (password: password123) - Regular User 3: ahmed.benali@example.com (password: password123)

Each user account includes realistic personal information, security settings, and preferences that demonstrate the full range of user profile capabilities. The accounts are configured with different language preferences (English, French, Arabic) to showcase internationalization support.

Financial Data: Each user is assigned a bank account with realistic account numbers following Algerian banking standards. Account balances range from 50,000 to 80,000 DZD, providing sufficient funds for testing various transaction scenarios. Daily and monthly transfer limits are set according to typical banking restrictions.

Transaction History: The system generates 20 sample transactions representing various types of banking operations including transfers between accounts, payments, and deposits. These transactions include both completed and pending statuses to demonstrate different transaction states and their handling within the application.

Notification Records: Each user receives sample notifications related to their account activity, including transaction alerts and system messages. These notifications demonstrate the notification system's capability to handle different message types and delivery statuses.

Database Performance and Indexing

Index Strategy: The database implementation includes strategic indexing to ensure optimal performance for common query patterns. Primary indexes are created on frequently accessed fields such as user email addresses, account numbers, and transaction dates.

Performance Considerations: The current schema is optimized for read-heavy operations typical in banking applications where users frequently check balances and transaction histories. Write operations are optimized through proper field selection and minimal document nesting where appropriate.

Scaling Considerations: The database design supports horizontal scaling through MongoDB's sharding capabilities. Key fields like `user_id` and `account_id` are suitable for shard key selection in high-volume production environments.

Database Maintenance and Backup

Regular Maintenance Tasks: Database maintenance should include regular index optimization, collection statistics updates, and performance monitoring. MongoDB provides built-in tools for these operations, and automated scripts can be developed for routine maintenance tasks.

Backup Strategy: Production deployments should implement comprehensive backup strategies including regular full backups, incremental backups, and point-in-time recovery capabilities. MongoDB Atlas provides automated backup solutions, while self-hosted deployments can use `mongodump` and `mongorestore` utilities.

Data Retention Policies: Banking applications require careful consideration of data retention policies for regulatory compliance. Transaction records should be retained according to local banking regulations, while user activity logs may have different retention requirements.

Backend API Documentation

The SATIM Pay backend provides a comprehensive RESTful API built with FastAPI, offering automatic documentation, request validation, and high-performance asynchronous operations. This section details the API architecture, endpoint specifications, and integration guidelines for frontend developers and third-party integrations.

API Architecture and Design Principles

RESTful Design: The API follows REST architectural principles with clear resource-based URLs, appropriate HTTP methods, and standardized response formats. Each endpoint is designed to perform specific operations on well-defined resources, ensuring predictable behavior and easy integration.

Asynchronous Operations: All API endpoints are implemented using Python's `async/await` pattern, providing excellent performance for I/O-bound operations such as database queries and external service calls. This architecture ensures the API can handle multiple concurrent requests efficiently.

Automatic Documentation: FastAPI automatically generates interactive API documentation available at `/docs` (Swagger UI) and `/redoc` (ReDoc). These interfaces provide comprehensive endpoint documentation, request/response schemas, and testing capabilities directly in the browser.

Authentication and Authorization

JWT Token-Based Authentication: The API uses JSON Web Tokens (JWT) for stateless authentication. Upon successful login, users receive an access token and refresh token. The access token must be included in the Authorization header for protected endpoints.

Token Management: Access tokens have a 30-minute expiration time for security, while refresh tokens are valid for 7 days. The frontend automatically handles token refresh to maintain user sessions without requiring frequent re-authentication.

Role-Based Access Control: The system supports multiple user roles (user, admin, support) with different permission levels. Administrative endpoints require admin role verification, while user-specific operations are restricted to the account owner or authorized personnel.

Core API Endpoints

Authentication Endpoints:

The authentication system provides secure login, registration, and token management capabilities. The login endpoint accepts either email or phone number along with a password, returning JWT tokens upon successful authentication.

```
POST /api/v1/auth/login
Content-Type: application/x-www-form-urlencoded

username: admin@satimpay.com
password: admin123

Response:
{
  "access_token": "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9...",
  "refresh_token": "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9...",
  "token_type": "bearer",
  "user": {
    "id": "user_id",
    "email": "admin@satimpay.com",
    "role": "admin",
    "first_name": "Admin",
    "last_name": "User"
  }
}
```

User Management Endpoints:

User management endpoints provide comprehensive user account operations including profile updates, security settings, and account status management.

```
GET /api/v1/auth/me
Authorization: Bearer {access_token}

Response:
{
  "id": "user_id",
  "email": "admin@satimpay.com",
  "first_name": "Admin",
  "last_name": "User",
  "role": "admin",
  "is_active": true,
  "created_at": "2025-06-21T01:00:00Z",
  "preferences": {
    "language": "en",
    "currency": "DZD",
    "notifications": true
  }
}
```

Transaction Endpoints:

Transaction endpoints handle all financial operations including transfers, payments, and transaction history retrieval. These endpoints include comprehensive validation and security checks.


```
GET /api/v1/transactions?page=1&limit=10
Authorization: Bearer {access_token}
```

Response:

```
{
  "transactions": [
    {
      "id": "transaction_id",
      "amount": 1500.00,
      "currency": "DZD",
      "type": "transfer",
      "status": "completed",
      "description": "Payment #1",
      "created_at": "2025-06-21T01:00:00Z",
      "from_account": "account_id_1",
      "to_account": "account_id_2"
    }
  ],
  "total": 20,
  "page": 1,
  "limit": 10
}
```

AI Chatbot Endpoints:

The AI chatbot integration provides intelligent customer support through natural language processing. The chatbot endpoints handle user queries and return contextually appropriate responses.

```
POST /api/v1/ai/chat
Authorization: Bearer {access_token}
Content-Type: application/json
```

```
{
  "message": "Comment puis-je consulter mon solde ?",
  "session_id": "optional_session_id"
}
```

Response:

```
{
  "response": "Pour consulter votre solde, connectez-vous à votre compte...",
  "session_id": "session_id",
  "suggestions": [
    "Voir mon solde",
    "Faire un virement",
    "Historique des transactions",
    "Aide et support"
  ]
}
```

Error Handling and Response Formats

Standardized Error Responses: All API endpoints return consistent error response formats with appropriate HTTP status codes and detailed error messages. This standardization simplifies error handling in frontend applications.

```
// Error Response Format
{
  "detail": "Error description",
  "error_code": "SPECIFIC_ERROR_CODE",
  "timestamp": "2025-06-21T01:00:00Z"
}
```

HTTP Status Codes: The API uses standard HTTP status codes consistently across all endpoints: - 200: Successful operation - 201: Resource created successfully - 400: Bad request (validation errors) - 401: Unauthorized (authentication required) - 403: Forbidden (insufficient permissions) - 404: Resource not found - 500: Internal server error

API Security Considerations

Input Validation: All API endpoints include comprehensive input validation using Pydantic models. This ensures data integrity and prevents common security vulnerabilities such as injection attacks.

Rate Limiting: Production deployments should implement rate limiting to prevent abuse and ensure fair resource allocation among users. FastAPI supports various rate limiting strategies through middleware.

CORS Configuration: The API is configured with appropriate CORS settings to allow frontend access while maintaining security. Production deployments should restrict CORS origins to specific domains.

Request Logging: All API requests are logged with appropriate detail levels for security monitoring and debugging purposes. Sensitive information such as passwords and tokens are excluded from logs.

Frontend Configuration

The SATIM Pay frontend is built using Next.js 15.2.4 with TypeScript, providing a modern, responsive, and type-safe user interface. This section covers the frontend architecture, component structure, configuration options, and customization guidelines for developers working on the user interface.

Frontend Architecture Overview

Next.js App Router: The application uses Next.js 15's App Router architecture, which provides improved performance, better developer experience, and enhanced routing capabilities. The App Router enables server-side rendering, static site generation, and client-side navigation with automatic code splitting.

TypeScript Integration: The entire frontend codebase is written in TypeScript, providing compile-time type checking, improved IDE support, and better code maintainability. Type definitions are provided for all API responses, component props, and application state.

Component Architecture: The frontend follows a component-based architecture with reusable UI components, page-specific components, and utility functions organized in a logical directory structure. This approach promotes code reusability and maintainability.

Project Structure and Organization

Directory Structure:

```

frontend/
├── app/                                # Next.js App Router pages
│   ├── auth/                          # Authentication pages
│   ├── dashboard/                     # User dashboard pages
│   ├── admin/                         # Administrative interface
│   ├── layout.tsx                     # Root layout component
│   └── page.tsx                       # Homepage component
├── components/                        # Reusable UI components
│   ├── ui/                           # Base UI components (buttons, inputs, etc.)
│   ├── floating-chatbot.tsx
│   └── [other-components]
├── lib/                               # Utility functions and configurations
│   ├── api.ts                        # API client and helper functions
│   └── utils.ts                      # General utility functions
├── public/                            # Static assets
│   ├── satim-logo.png
│   └── [other-assets]
├── styles/                            # Global styles
│   └── globals.css
├── package.json                      # Dependencies and scripts
├── tailwind.config.ts                # Tailwind CSS configuration
├── tsconfig.json                     # TypeScript configuration
└── .env.local                        # Environment variables

```

Component Organization: Components are organized by functionality and reusability. Base UI components in the `components/ui/` directory provide consistent styling and behavior across the application. Page-specific components are located within their respective page directories in the `app/` folder.

Styling and Design System

Tailwind CSS Integration: The application uses Tailwind CSS for utility-first styling, providing rapid development capabilities and consistent design patterns. The Tailwind configuration is customized to match SATIM Pay's brand colors and design requirements.

Brand Color Palette:

```

/* Primary SATIM Red */
--satim-red: #E2211C;
--satim-red-dark: #C11E18;
--satim-red-light: #B01A15;

/* Supporting Colors */
--background: #FFFFFF;
--foreground: #1F2937;
--muted: #F3F4F6;
--border: #E5E7EB;

```

Radix UI Components: The application leverages Radix UI components for accessible, unstyled UI primitives. These components provide excellent accessibility features and consistent behavior across different browsers and devices.

Responsive Design: All components are designed with mobile-first responsive principles, ensuring optimal user experience across desktop, tablet, and mobile devices. The design adapts seamlessly to different screen sizes using Tailwind's responsive utilities.

State Management and Data Flow

API Client Integration: The frontend includes a comprehensive API client (`lib/api.ts`) that handles all communication with the backend. The client provides methods for authentication, user management, transactions, and AI chatbot interactions.

Authentication State: User authentication state is managed through `localStorage` for persistence and React state for real-time updates. The API client automatically includes authentication tokens in requests and handles token refresh operations.

Form Management: Forms throughout the application use React Hook Form with Zod schema validation, providing type-safe form handling with excellent user experience and performance characteristics.

Key Frontend Features

Homepage and Landing Page: The homepage showcases SATIM Pay's features with a modern, professional design that emphasizes security, innovation, and user experience. The page includes interactive elements, feature highlights, and clear calls-to-action for user registration and login.

Authentication Interface: The authentication system provides both traditional password-based login and biometric authentication options. The interface is designed for accessibility and includes features such as password visibility toggles, remember me functionality, and forgot password links.

AI Chatbot Integration: The floating chatbot component provides seamless AI-powered customer support. The chatbot interface includes message history, quick action buttons, and real-time communication with the backend AI service.

Dashboard Components: User dashboard components provide comprehensive account management capabilities including balance displays, transaction history, transfer functionality, and notification management.

Environment Configuration

Environment Variables: The frontend uses environment variables for configuration management, allowing different settings for development, staging, and production environments.

```
# .env.local
NEXT_PUBLIC_API_URL=http://localhost:8000
NEXT_PUBLIC_APP_NAME=SATIM Pay
NEXT_PUBLIC_VERSION=1.0.0
```

Build Configuration: The Next.js configuration supports various build optimizations including image optimization, bundle analysis, and performance monitoring. The configuration can be customized in `next.config.mjs` for specific deployment requirements.

Development and Build Process

Development Server: The development server provides hot reloading, error overlay, and development-specific optimizations for rapid development cycles.

```
# Start development server
npm run dev

# Build for production
npm run build

# Start production server
npm start

# Run linting
npm run lint
```

Code Quality Tools: The project includes ESLint configuration for code quality enforcement and TypeScript for compile-time error checking. These tools help maintain code consistency and catch potential issues early in the development process.

Customization and Extension

Theme Customization: The application's visual theme can be customized through the Tailwind configuration file. Brand colors, typography, spacing, and other design tokens can be modified to match specific branding requirements.

Component Extension: New components can be added following the existing patterns and conventions. The component library is designed for extensibility, allowing developers to create new features while maintaining consistency with the existing design system.

Internationalization Support: The application is structured to support internationalization (i18n) with language-specific content and formatting. Additional languages can be added by extending the existing language configuration and providing translated content.

Authentication and Security

Security is paramount in banking applications, and the SATIM Pay system implements multiple layers of security controls to protect user data, prevent unauthorized access, and ensure compliance with banking security standards. This section details the comprehensive security architecture and implementation guidelines.

Authentication Architecture

Multi-Factor Authentication System: The application supports multiple authentication methods including traditional password-based authentication and biometric authentication. This multi-layered approach provides users with secure and convenient access options while maintaining high security standards.

JWT Token Management: The system uses JSON Web Tokens (JWT) for stateless authentication, providing scalable session management without server-side session storage. Access tokens are short-lived (30 minutes) to minimize security exposure, while refresh tokens allow seamless session continuation for up to 7 days.

Password Security: User passwords are hashed using bcrypt with a configurable number of rounds (default: 12) to provide strong protection against rainbow table and

brute-force attacks. The system enforces password complexity requirements and prevents the use of common or compromised passwords.

Security Implementation Details

Password Hashing and Validation:

```
# Password hashing implementation
from passlib.context import CryptContext

pwd_context = CryptContext(schemes=["bcrypt"], deprecated="auto")

def hash_password(password: str) -> str:
    return pwd_context.hash(password)

def verify_password(plain_password: str, hashed_password: str) -> bool:
    return pwd_context.verify(plain_password, hashed_password)
```

JWT Token Generation:

```
# JWT token implementation
from jose import JWTError, jwt
from datetime import datetime, timedelta

def create_access_token(data: dict, expires_delta: timedelta = None):
    to_encode = data.copy()
    if expires_delta:
        expire = datetime.utcnow() + expires_delta
    else:
        expire = datetime.utcnow() + timedelta(minutes=30)

    to_encode.update({"exp": expire})
    encoded_jwt = jwt.encode(to_encode, SECRET_KEY, algorithm=ALGORITHM)
    return encoded_jwt
```

Account Security Features: The system includes comprehensive account security features such as login attempt monitoring, account lockout mechanisms, and suspicious activity detection. Failed login attempts are tracked and accounts are temporarily locked after exceeding the maximum allowed attempts.

Data Protection and Privacy

Data Encryption: All sensitive data is encrypted both in transit and at rest. HTTPS is enforced for all communications, and database connections use encrypted channels. Personal and financial information is encrypted using industry-standard algorithms.

Input Validation and Sanitization: All user inputs are validated and sanitized to prevent injection attacks, cross-site scripting (XSS), and other common web vulnerabilities. The FastAPI framework provides automatic request validation using Pydantic models.

CORS and Security Headers: The application implements appropriate Cross-Origin Resource Sharing (CORS) policies and security headers to prevent unauthorized access and protect against common web attacks.

Compliance and Regulatory Considerations

Banking Security Standards: The application is designed to meet banking industry security standards including PCI DSS requirements for payment processing and local banking regulations in Algeria. Security controls are implemented according to these standards.

Audit Trail and Logging: Comprehensive audit trails are maintained for all user actions, authentication events, and system operations. These logs are essential for security monitoring, incident response, and regulatory compliance.

Data Retention and Privacy: The system implements appropriate data retention policies and privacy controls in accordance with banking regulations and data protection laws. Users have control over their personal data and can request data deletion where legally permitted.

Security Monitoring and Incident Response

Real-time Security Monitoring: The application includes security monitoring capabilities to detect and respond to potential threats in real-time. This includes monitoring for unusual login patterns, suspicious transactions, and system anomalies.

Incident Response Procedures: Clear incident response procedures are established for handling security incidents, including data breaches, unauthorized access attempts, and system compromises. These procedures ensure rapid response and minimize potential damage.

Security Updates and Maintenance: Regular security updates and maintenance procedures are essential for maintaining the security posture of the application. This includes updating dependencies, applying security patches, and reviewing security configurations.

AI Chatbot Integration

The SATIM Pay application features an advanced AI-powered chatbot that provides intelligent customer support and banking assistance. The chatbot is integrated using the Groq API and provides contextual responses in French, Arabic, and English to serve the diverse Algerian banking market.

Chatbot Architecture and Design

AI Service Integration: The chatbot leverages the Groq API for natural language processing and response generation. Groq provides high-performance AI inference with low latency, making it ideal for real-time customer support applications.

Multilingual Support: The chatbot is designed to handle queries in multiple languages, primarily French and Arabic, with English support for international users. The AI model is trained to understand banking terminology and provide appropriate responses in the user's preferred language.

Context Awareness: The chatbot maintains conversation context and can provide personalized responses based on user account information and transaction history. This context awareness enables more helpful and relevant assistance.

Implementation Details

Backend Integration: The chatbot functionality is implemented through dedicated API endpoints that process user messages and return AI-generated responses. The backend handles message preprocessing, AI service communication, and response formatting.

```

# Chatbot service implementation
async def process_chat_message(message: str, user_id: str = None) -> str:
    # Preprocess message for banking context
    banking_context = f"Tu es un assistant bancaire pour SATIM Pay en Algérie. Réponds en français: {message}"

    # Call Groq AI service
    response = await groq_client.chat.completions.create(
        messages=[{"role": "user", "content": banking_context}],
        model="mixtral-8x7b-32768",
        max_tokens=500
    )

    return response.choices[0].message.content

```

Frontend Integration: The frontend chatbot component provides an intuitive interface with message history, typing indicators, and quick action buttons for common banking operations.

```

// Chatbot API integration
const handleMessage = async (message: string) => {
    try {
        const response = await apiClient.chatWithBot(message);
        if (response.data?.response) {
            // Display AI response
            addBotMessage(response.data.response);
        }
    } catch (error) {
        // Handle error gracefully
        addBotMessage("Je rencontre des difficultés techniques. Veuillez réessayer.");
    }
};

```

Chatbot Features and Capabilities

Banking Knowledge Base: The chatbot is equipped with comprehensive knowledge about banking operations, SATIM Pay features, and common customer inquiries. It can provide information about account balances, transaction procedures, security features, and general banking assistance.

Quick Actions: The chatbot interface includes quick action buttons for common operations such as checking balances, making payments, viewing transaction history, and accessing account settings. These shortcuts improve user experience and reduce conversation complexity.

Conversation Management: The system maintains conversation sessions and can handle follow-up questions and context-dependent queries. Session management

ensures continuity in multi-turn conversations.

Customization and Enhancement

Response Customization: The chatbot responses can be customized to match SATIM Pay's brand voice and communication style. Response templates and conversation flows can be modified to improve user experience.

Knowledge Base Expansion: The chatbot's knowledge base can be expanded with additional banking information, frequently asked questions, and specific SATIM Pay procedures. This expansion improves the chatbot's ability to handle diverse customer inquiries.

Analytics and Improvement: Chatbot interactions can be analyzed to identify common user queries, conversation patterns, and areas for improvement. This data helps optimize the chatbot's performance and user satisfaction.

Testing and Validation

Comprehensive testing is essential for ensuring the reliability, security, and performance of the SATIM Pay banking application. This section outlines the testing strategies, validation procedures, and quality assurance processes implemented throughout the development lifecycle.

Testing Strategy Overview

Multi-Layer Testing Approach: The application employs a comprehensive testing strategy that includes unit testing, integration testing, end-to-end testing, and security testing. This multi-layer approach ensures that all components function correctly both individually and as part of the complete system.

Automated Testing Pipeline: Automated testing is integrated into the development workflow to catch issues early and maintain code quality. The testing pipeline includes code quality checks, security scans, and functional testing across different environments.

Manual Testing Procedures: While automated testing covers the majority of test cases, manual testing is performed for user experience validation, accessibility testing,

and complex user workflows that require human judgment.

Backend Testing Implementation

Unit Testing Framework: The backend uses pytest for unit testing, providing comprehensive test coverage for individual functions, API endpoints, and business logic components. Tests are organized by functionality and include both positive and negative test cases.

```
# Example unit test for authentication
import pytest
from app.services.auth_service import AuthService

@pytest.mark.asyncio
async def test_user_authentication():
    auth_service = AuthService()

    # Test successful authentication
    result = await auth_service.authenticate_user(
        "admin@satimpay.com",
        "admin123"
    )
    assert result is not None
    assert result.email == "admin@satimpay.com"

    # Test failed authentication
    result = await auth_service.authenticate_user(
        "admin@satimpay.com",
        "wrong_password"
    )
    assert result is None
```

API Integration Testing: Integration tests verify that API endpoints function correctly with the database and external services. These tests use test databases to ensure data isolation and repeatability.

```

# Example API integration test
@pytest.mark.asyncio
async def test_transaction_creation():
    async with AsyncClient(app=app, base_url="http://test") as client:
        # Login to get authentication token
        login_response = await client.post("/api/v1/auth/login", data={
            "username": "admin@satimpay.com",
            "password": "admin123"
        })
        token = login_response.json()["access_token"]

        # Create transaction
        transaction_response = await client.post(
            "/api/v1/transactions/transfer",
            headers={"Authorization": f"Bearer {token}"},
            json={
                "to_account_id": "account_id",
                "amount": 1000.0,
                "description": "Test transfer"
            }
        )
        assert transaction_response.status_code == 201

```

Database Testing: Database tests verify data integrity, constraint enforcement, and query performance. These tests use MongoDB test instances to ensure isolation from production data.

Frontend Testing Implementation

Component Testing: Frontend components are tested using Jest and React Testing Library, focusing on component behavior, user interactions, and integration with the API client.

```

// Example component test
import { render, screen, fireEvent } from '@testing-library/react';
import { FloatingChatbot } from '@components/floating-chatbot';

describe('FloatingChatbot', () => {
  test('opens chatbot when button is clicked', () => {
    render(<FloatingChatbot />);

    const chatButton = screen.getByRole('button');
    fireEvent.click(chatButton);

    expect(screen.getByText('Assistant SATIM')).toBeInTheDocument();
  });

  test('sends message when form is submitted', async () => {
    render(<FloatingChatbot />);

    // Open chatbot and send message
    fireEvent.click(screen.getByRole('button'));

    const input = screen.getByPlaceholderText('Tapez votre message...');
    fireEvent.change(input, { target: { value: 'Test message' } });
    fireEvent.click(screen.getByText('Send'));

    expect(screen.getByText('Test message')).toBeInTheDocument();
  });
});

```

End-to-End Testing: End-to-end tests simulate complete user workflows using tools like Playwright or Cypress, ensuring that the entire application functions correctly from the user's perspective.

```

// Example E2E test
import { test, expect } from '@playwright/test';

test('user can login and view dashboard', async ({ page }) => {
  await page.goto('http://localhost:3000');

  // Navigate to login page
  await page.click('text=Connexion');

  // Fill login form
  await page.fill('input[placeholder*="email"]', 'admin@satimpay.com');
  await page.fill('input[placeholder*="password"]', 'admin123');
  await page.click('button:has-text("Se connecter)");

  // Verify dashboard loads
  await expect(page).toHaveURL(/.*dashboard/);
  await expect(page.locator('text=Tableau de bord')).toBeVisible();
});

```

Security Testing Procedures

Authentication Testing: Security tests verify that authentication mechanisms function correctly and prevent unauthorized access. These tests include password validation, token expiration, and session management.

Input Validation Testing: All user inputs are tested for proper validation and sanitization to prevent injection attacks, XSS vulnerabilities, and other security issues.

Authorization Testing: Tests verify that users can only access resources and perform actions appropriate to their role and permissions.

Performance Testing

Load Testing: Load tests simulate multiple concurrent users to verify that the application performs adequately under expected traffic loads. These tests identify performance bottlenecks and scalability limitations.

Database Performance Testing: Database queries are tested for performance with realistic data volumes to ensure acceptable response times for common operations.

API Response Time Testing: API endpoints are tested for response time performance to ensure they meet user experience requirements.

Test Data Management

Test Database Setup: Separate test databases are used for different testing scenarios, ensuring data isolation and repeatability. Test data is automatically created and cleaned up for each test run.

Dummy Data Generation: Realistic test data is generated to simulate production scenarios while maintaining data privacy and security. The test data includes various user types, account configurations, and transaction patterns.

Data Privacy in Testing: All test data is anonymized and does not contain real personal or financial information. Production data is never used in testing environments.

Continuous Integration and Quality Assurance

Automated Test Execution: Tests are automatically executed as part of the continuous integration pipeline, ensuring that code changes do not introduce regressions or break existing functionality.

Code Coverage Monitoring: Code coverage metrics are tracked to ensure comprehensive test coverage across all application components. Minimum coverage thresholds are enforced to maintain quality standards.

Quality Gates: Quality gates are implemented to prevent deployment of code that does not meet testing and quality standards. These gates include test pass rates, code coverage thresholds, and security scan results.

Deployment Guidelines

Deploying the SATIM Pay application requires careful consideration of infrastructure requirements, security configurations, and operational procedures. This section provides comprehensive guidelines for deploying the application in various environments from development to production.

Deployment Architecture Options

Single Server Deployment: For development and small-scale testing environments, the entire application stack can be deployed on a single server. This approach simplifies initial setup and is suitable for proof-of-concept deployments.

Multi-Server Architecture: Production deployments should use a multi-server architecture with separate servers for the frontend, backend API, and database components. This separation provides better performance, security, and scalability.

Cloud-Native Deployment: The application is designed to support cloud-native deployment using containerization and orchestration platforms such as Docker and Kubernetes. This approach provides excellent scalability and operational flexibility.

Microservices Architecture: For large-scale deployments, the application can be decomposed into microservices with separate deployment units for authentication, transaction processing, notification services, and AI chatbot functionality.

Infrastructure Requirements

Server Specifications: Production servers should meet minimum hardware requirements to ensure adequate performance and reliability. The following specifications are recommended for different deployment scenarios:

Frontend Server Requirements: - CPU: 2+ cores - RAM: 4GB minimum, 8GB recommended - Storage: 20GB SSD - Network: 1Gbps connection - Operating System: Ubuntu 22.04 LTS or equivalent

Backend Server Requirements: - CPU: 4+ cores - RAM: 8GB minimum, 16GB recommended - Storage: 50GB SSD - Network: 1Gbps connection - Operating System: Ubuntu 22.04 LTS or equivalent

Database Server Requirements: - CPU: 4+ cores - RAM: 16GB minimum, 32GB recommended - Storage: 100GB SSD with backup storage - Network: 1Gbps connection - Operating System: Ubuntu 22.04 LTS or equivalent

Production Deployment Process

Environment Preparation: Production environments require careful preparation including server provisioning, network configuration, security hardening, and monitoring setup.

```
# Server preparation script
#!/bin/bash

# Update system packages
sudo apt update && sudo apt upgrade -y

# Install required software
sudo apt install -y nginx certbot python3-certbot-nginx
sudo apt install -y nodejs npm mongodb-org
sudo apt install -y python3 python3-pip python3-venv

# Configure firewall
sudo ufw allow ssh
sudo ufw allow http
sudo ufw allow https
sudo ufw enable

# Create application user
sudo useradd -m -s /bin/bash satimpay
sudo usermod -aG sudo satimpay
```

Application Deployment: The deployment process involves building the application, configuring services, and starting all components in the correct order.

```
# Backend deployment
cd /opt/satimpay/backend
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt

# Configure environment variables
cp .env.example .env
nano .env # Edit production settings

# Start backend service
gunicorn app.main:app -w 4 -k uvicorn.workers.UvicornWorker --bind 0.0.0.0:8000

# Frontend deployment
cd /opt/satimpay/frontend
npm ci --production
npm run build

# Configure Nginx
sudo cp nginx.conf /etc/nginx/sites-available/satimpay
sudo ln -s /etc/nginx/sites-available/satimpay /etc/nginx/sites-enabled/
sudo nginx -t && sudo systemctl reload nginx
```

SSL Certificate Configuration: Production deployments must use SSL/TLS encryption for all communications. Let's Encrypt provides free SSL certificates that can be automatically configured.

```
# SSL certificate setup
sudo certbot --nginx -d satimpay.dz -d www.satimpay.dz
sudo systemctl enable certbot.timer
```

Database Deployment and Configuration

MongoDB Production Setup: Production MongoDB deployments require specific configuration for security, performance, and reliability.

```
// MongoDB production configuration
{
  "storage": {
    "dbPath": "/var/lib/mongodb",
    "journal": {
      "enabled": true
    }
  },
  "systemLog": {
    "destination": "file",
    "logAppend": true,
    "path": "/var/log/mongodb/mongod.log"
  },
  "net": {
    "port": 27017,
    "bindIp": "127.0.0.1"
  },
  "security": {
    "authorization": "enabled"
  },
  "replication": {
    "replSetName": "satimpay-rs"
  }
}
```

Database Security Configuration: Production databases must be secured with authentication, authorization, and network access controls.

```
# MongoDB security setup
mongo admin --eval "
  db.createUser({
    user: 'admin',
    pwd: 'secure_admin_password',
    roles: ['userAdminAnyDatabase', 'dbAdminAnyDatabase']
  })
"

mongo satim_pay --eval "
  db.createUser({
    user: 'satimpay_app',
    pwd: 'secure_app_password',
    roles: ['readWrite']
  })
"
```

Backup and Recovery Setup: Automated backup procedures are essential for production deployments to ensure data protection and disaster recovery capabilities.

```

# Automated backup script
#!/bin/bash
BACKUP_DIR="/backup/mongodb"
DATE=$(date +%Y%m%d_%H%M%S)

# Create backup
mongodump --host localhost:27017 --db satim_pay --out `${BACKUP_DIR}/${DATE}`

# Compress backup
tar -czf `${BACKUP_DIR}/satim_pay_${DATE}.tar.gz` `${BACKUP_DIR}/${DATE}`
rm -rf `${BACKUP_DIR}/${DATE}`

# Upload to cloud storage (optional)
aws s3 cp `${BACKUP_DIR}/satim_pay_${DATE}.tar.gz` s3://satimpay-backups/

# Cleanup old backups (keep last 30 days)
find $BACKUP_DIR -name "*.tar.gz" -mtime +30 -delete

```

Monitoring and Logging

Application Monitoring: Production deployments require comprehensive monitoring to track application performance, availability, and user experience.

Log Management: Centralized log management is essential for troubleshooting, security monitoring, and compliance requirements.

```

# Log rotation configuration
sudo tee /etc/logrotate.d/satimpay << EOF
/var/log/satimpay/*.log {
    daily
    missingok
    rotate 30
    compress
    delaycompress
    notifempty
    create 644 satimpay satimpay
    postrotate
        systemctl reload satimpay-backend
        systemctl reload satimpay-frontend
    endscript
}
EOF

```

Health Checks and Alerting: Automated health checks and alerting systems ensure rapid response to system issues and minimize downtime.

Security Hardening

Server Security: Production servers must be hardened against security threats through proper configuration, access controls, and monitoring.

Application Security: Application-level security measures include input validation, authentication controls, and secure communication protocols.

Network Security: Network security controls include firewalls, intrusion detection systems, and secure communication channels.

Troubleshooting

This section provides comprehensive troubleshooting guidance for common issues that may arise during development, deployment, and operation of the SATIM Pay application. The troubleshooting procedures are organized by component and include diagnostic steps and resolution strategies.

Common Installation Issues

Python Dependency Conflicts: Python dependency conflicts can occur when installing packages with incompatible version requirements. These issues are typically resolved by using virtual environments and updating package versions.

```
# Resolve Python dependency conflicts
cd backend
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt --force-reinstall
```

Node.js Package Installation Errors: Frontend package installation may fail due to peer dependency conflicts or version incompatibilities. The `--legacy-peer-deps` flag resolves most compatibility issues.

```
# Resolve Node.js package conflicts
cd frontend
rm -rf node_modules package-lock.json
npm cache clean --force
npm install --legacy-peer-deps
```

MongoDB Connection Issues: Database connection problems can result from incorrect configuration, network issues, or authentication failures.

```
# Diagnose MongoDB connection
# Check MongoDB service status
sudo systemctl status mongod

# Test database connection
mongo --eval "db.adminCommand('ping')"

# Check MongoDB logs
sudo tail -f /var/log/mongodb/mongod.log
```

Backend API Issues

Server Startup Failures: Backend server startup failures are often caused by port conflicts, missing dependencies, or configuration errors.

```
# Diagnose backend startup issues
cd backend

# Check for port conflicts
sudo netstat -tulpn | grep :8000

# Test configuration
python3 -c "from app.core.config import settings; print(settings.MONGODB_URL)"

# Run with verbose logging
python3 standalone_main.py --log-level debug
```

Database Connection Errors: Database connection errors can result from incorrect connection strings, authentication failures, or network connectivity issues.

```
# Test database connection
import asyncio
from motor.motor_asyncio import AsyncIOMotorClient

async def test_connection():
    client = AsyncIOMotorClient("mongodb://localhost:27017")
    try:
        await client.admin.command('ping')
        print("Database connection successful")
    except Exception as e:
        print(f"Database connection failed: {e}")
    finally:
        client.close()

asyncio.run(test_connection())
```

API Authentication Issues: Authentication problems can occur due to incorrect token handling, expired tokens, or configuration errors.

```
# Test API authentication
curl -X POST http://localhost:8000/api/v1/auth/login \
  -H "Content-Type: application/x-www-form-urlencoded" \
  -d "username=admin@satimpay.com&password=admin123"

# Test protected endpoint
curl -X GET http://localhost:8000/api/v1/auth/me \
  -H "Authorization: Bearer YOUR_TOKEN_HERE"
```

Frontend Issues

Build and Compilation Errors: Frontend build errors are typically caused by TypeScript type errors, missing dependencies, or configuration issues.

```
# Diagnose frontend build issues
cd frontend

# Check TypeScript compilation
npx tsc --noEmit

# Clear Next.js cache
rm -rf .next

# Rebuild with verbose output
npm run build -- --debug
```

API Communication Errors: Frontend-backend communication issues can result from CORS configuration, network connectivity, or API endpoint changes.

```
// Test API connectivity
const testApiConnection = async () => {
  try {
    const response = await fetch('http://localhost:8000/health');
    const data = await response.json();
    console.log('API connection successful:', data);
  } catch (error) {
    console.error('API connection failed:', error);
  }
};
```

Chatbot Integration Issues: Chatbot functionality may fail due to API key configuration, network connectivity, or service availability.


```
# Test Groq API connectivity
curl -X POST https://api.groq.com/openai/v1/chat/completions \
  -H "Authorization: Bearer YOUR_GROQ_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{
    "messages": [{"role": "user", "content": "Test message"}],
    "model": "mixtral-8x7b-32768"
  }'
```

Database Issues

Performance Problems: Database performance issues can result from missing indexes, inefficient queries, or insufficient resources.

```
// Check database performance
db.users.explain("executionStats").find({"email": "admin@satimpay.com"})

// Create missing indexes
db.users.createIndex({"email": 1}, {"unique": true})
db.transactions.createIndex({"created_at": -1})
db.accounts.createIndex({"user_id": 1})
```

Data Integrity Issues: Data integrity problems can occur due to application bugs, concurrent access, or manual data modifications.

```
// Validate data integrity
// Check for orphaned records
db.accounts.find({"user_id": {$nin: db.users.distinct("_id")}})

// Verify account balances
db.accounts.aggregate([
  {$lookup: {
    from: "transactions",
    localField: "_id",
    foreignField: "from_account",
    as: "outgoing"
  }},
  {$lookup: {
    from: "transactions",
    localField: "_id",
    foreignField: "to_account",
    as: "incoming"
  }}
])
```

Production Deployment Issues

SSL Certificate Problems: SSL certificate issues can prevent secure connections and cause browser warnings.

```
# Check SSL certificate status
sudo certbot certificates

# Renew SSL certificates
sudo certbot renew --dry-run

# Test SSL configuration
openssl s_client -connect satimpay.dz:443 -servername satimpay.dz
```

Load Balancer Configuration: Load balancer misconfigurations can cause service unavailability or poor performance.

```
# Check Nginx configuration
sudo nginx -t

# Monitor Nginx access logs
sudo tail -f /var/log/nginx/access.log

# Check upstream server health
curl -I http://localhost:8000/health
```

Resource Exhaustion: Resource exhaustion can cause application failures and poor performance.

```
# Monitor system resources
htop
df -h
free -m

# Check application resource usage
ps aux | grep python
ps aux | grep node

# Monitor database resources
mongo --eval "db.serverStatus().mem"
```

Diagnostic Tools and Procedures

Log Analysis: Comprehensive log analysis is essential for identifying and resolving issues.

```
# Application logs
tail -f /var/log/satipay/backend.log
tail -f /var/log/satipay/frontend.log

# System logs
sudo journalctl -u satipay-backend -f
sudo journalctl -u satipay-frontend -f

# Database logs
sudo tail -f /var/log/mongodb/mongod.log
```

Performance Monitoring: Performance monitoring tools help identify bottlenecks and optimization opportunities.

```
# Monitor API response times
curl -w "@curl-format.txt" -o /dev/null -s http://localhost:8000/api/v1/auth/me

# Database performance monitoring
mongostat --host localhost:27017
mongotop --host localhost:27017
```

Network Diagnostics: Network diagnostic tools help identify connectivity and communication issues.

```
# Test network connectivity
ping satipay.dz
telnet localhost 8000
nslookup satipay.dz

# Monitor network traffic
sudo netstat -tulpn
sudo ss -tulpn
```

Conclusion

The SATIM Pay full-stack web application represents a comprehensive banking platform built with modern technologies and best practices. This documentation has provided detailed guidance for developers to understand, deploy, and maintain the application across various environments and use cases.

The successful integration of all components demonstrates the application's readiness for production deployment with proper security measures, performance optimizations, and operational procedures. The modular architecture ensures

scalability and maintainability while the comprehensive testing and documentation support long-term development and operations.

Developers can use this documentation as a complete reference for working with the SATIM Pay application, from initial setup through advanced customization and troubleshooting. The application's design supports future enhancements and integrations while maintaining security and performance standards appropriate for banking applications.

For additional support or questions not covered in this documentation, developers should refer to the project's issue tracking system or contact the development team through established communication channels.

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